

Introduction to Naive Bayes for Classification

What does Naive and Bayes signify?

- This algorithm is called “Naive” because it makes a naive assumption that each feature is independent of other features which is not true in real life.
- As for the “Bayes” part, it refers to the statistician and philosopher, Thomas Bayes and the theorem named after him, Bayes’ theorem, which is the base for Naive Bayes Algorithm.

What is Naive Bayes Algorithm?

- Naive Bayes algorithm can be defined as a supervised classification algorithm which is based on **Bayes theorem** with an assumption of **independence among features**.

What is Conditional Probability?

- When you flip a fair coin, there is an equal chance of getting either heads or tails. So you can say the probability of getting heads is 50%.
- Similarly what would be the probability of getting a 1 when you roll a dice with 6 faces? Assuming the dice is fair, the probability of $1/6 = 0.166$.
- If you pick a card from the deck, can you guess the probability of getting a queen given the card is a spade?
- As there is only one queen in spades, the probability it is a queen given the card is a spade is $1/13 = 0.077$

What is Conditional Probability?

- So, when you say the conditional probability of A given B, it denotes the probability of A occurring given that B has already occurred.
- Mathematically, Conditional probability of A given B can be computed as: $P(A|B) = P(A \text{ AND } B) / P(B)$

Example

- With below tabulation of the 100 people, what is the conditional probability that a certain member of the school is a 'Teacher' given that he is a 'Man'?
- To calculate this, you may intuitively filter the sub-population of 60 males and focus on the 12 (male) teachers.
- So the required conditional probability $P(\text{Teacher} \mid \text{Male}) = 12 / 60 = 0.2$

	Female	Male	Total
Teacher	8	12	20
Student	32	48	80
Total	40	60	100

$$P(\text{Teacher} \mid \text{Male}) = \frac{P(\text{Teacher} \cap \text{Male})}{P(\text{Male})} = 12/60 = 0.2$$

Conditional Probability

- This can be represented as the intersection of Teacher (A) and Male (B) divided by Male (B). Likewise, the conditional probability of B given A can be computed. The Bayes Rule that we use for Naive Bayes, can be derived from these two notations.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad (1)$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \quad (2)$$

The Bayes Rule

Bayes Rule is a way to go from $P(X | Y)$ to find $P(Y | X)$

$$\underset{\text{Known}}{P(X | Y)} = \frac{P(X \cap Y)}{P(Y)} \quad 1$$

$P(\text{Evidence} | \text{Outcome})$
(Known from training data)

$$\underset{\text{UnKnown}}{P(Y | X)} = \frac{P(X \cap Y)}{P(X)} \quad 2$$

$P(\text{Outcome} | \text{Evidence})$
(To be predicted for test data)



Bayes Rule

$$P(Y | X) = \frac{P(X | Y) * P(Y)}{P(X)}$$

Naïve Bayes Analysis

- Naive-Bayes (NB) technique is a supervised learning technique that uses probability-theory-based analysis.
- It is a machine-learning technique that computes the probabilities of an instance belonging to each one of many target classes, given the prior probabilities of classification using individual factors.
- Naïve Bayes technique is used often in classifying text documents into one of multiple predefined categories.

Naïve Bayes Analysis

- NB algorithm is easy to understand and works fast.
- It also performs well in multiclass prediction, such as when the target class has multiple options beyond binary yes/no classification.
- NB can perform well even in case of categorical input variables compared to numerical variable(s)

Naïve Bayes Model

- In the abstract, Naive-Bayes is a conditional probability model for classification purposes.
- The goal is to find a way to predict the class variable (Y) using a vector of independent variables i.e., finding the function $f: X \rightarrow Y$.
- In probability terms, the goal is to find $P(Y|X)$, i.e., the probability of Y belonging to a certain class X .
- Y is generally assumed to be a categorical variable with two or more discrete values.

Naïve Bayes Model

- Given an instance to be classified, represented by a vector $x = (x_1, x_2, \dots, x_n)$ representing 'n' features (independent variables), the Naive-Bayes model assigns, to an instance, probabilities of belonging to any of the K classes.
- The class K with the highest posterior probability is the label assigned to the instance.

- The posterior probability (of belonging to a Class K) is calculated as a function of prior probabilities and current likelihood value, as shown in the equation,

$$p(c_k|x) = \frac{p(c_k) * p(x|c_k)}{p(x)}$$

- $p(c_k|x)$ is the posterior probability of class K, given predictor X.
- $p(c_k)$ is the prior probability of class K.
- $p(x)$ is the prior probability of predictor.
- $p(x|c_k)$ is the current likelihood of predictor given class.

Simple Classification Example

- Suppose a **XYZ** company needs to predict the service required by the incoming customer.
- If there are only two services offered — R and M.
- Then the value to be predicted is whether the next customer will be for R or M.
- The number of classes k are 2 that is, $k=2$.

Simple Classification Example

- The first step is to compute prior probability.
- Suppose the data gathered for the last one year showed that during that period there were 2500 customers for R and 1500 customers for M.
- Thus, prior probability for the next customer to be for R is $2500/4000$ or $5/8$.
- Prior probability for the next customer to be for M is $1500/4000$ or $3/8$.
- Based on this information alone, the next customer would likely be for R.

Simple Classification Example

- Another way to predict the service requirement by the next customer is to look at the most recent data.
- One can look at the last few (choose a number) customers, to predict the next customer.
- Suppose the last five customers were for the services ... R, M, R, M, M order.
- Thus, the data shows the recent probability of R is $\frac{2}{5}$ and that of M is $\frac{3}{5}$.
- Based on just this information, the next customer will likely to be for M.

Simple Classification Example

- Thomas Bayes suggested that the prior probability should be informed by the more recent data.
- Naive-Bayes posterior probability for a class is computed by multiplying the prior probability and the recent probability.
- NB posterior probability $P(R)$ is $(5/8 \times 2/5) = 10/40$.
- Similarly, the NB probability $P(M)$ is $(3/8 \times 3/5) = 9/40$.
- Since $P(R)$ is greater than $P(M)$, it follows that there is a greater probability of the next customer to be for R.
- Thus the expected class label assigned to the next customer would be R.

Simple Classification Example

- Suppose, however the next customer coming in was for M service.
- The last five customer sequence now becomes M, R, M, M, M.
- Thus, the recent data shows the probability for R to be $1/5$ and that of M to be $4/5$.
- Now the NB probability for R is $(5/8 \times 1/5) = 5/40$.
- Similarly, the NB probability for M is $(3/8 \times 4/5) = 12/40$.
- Since $P(M)$ is greater than $P(R)$, it follows that there is a greater probability of the next customer to be for M.
- Thus the expected class label assigned to the next customer is M.

Naïve Bayes Classifier Algorithm

- Bayes theorem is the cornerstone of Naïve Bayes Classifier because it provides a way to calculate the posterior probability $P(h|D)$, from the prior probability $P(h)$, together with $P(D)$ and $P(D|h)$.

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

$$h_{MAP} \equiv \operatorname{argmax}_{h \in H} P(h|D)$$

$$= \operatorname{argmax}_{h \in H} \frac{P(D|h)P(h)}{P(D)}$$

$$= \operatorname{argmax}_{h \in H} P(D|h)P(h)$$

Naïve Bayes Classifier Algorithm

- The Bayesian approach to classifying the new instance is to assign the most probable target value, $\mathbf{v}_{MAP}$ given the attribute values $\langle \mathbf{a1}, \mathbf{a2} \dots \mathbf{an} \rangle$ that describe the instance.

$$v_{MAP} = \operatorname{argmax}_{v_j \in V} P(v_j | a_1, a_2 \dots a_n)$$

- We can use Bayes theorem to rewrite this expression as

$$\begin{aligned} v_{MAP} &= \operatorname{argmax}_{v_j \in V} \frac{P(a_1, a_2 \dots a_n | v_j) P(v_j)}{P(a_1, a_2 \dots a_n)} \\ &= \operatorname{argmax}_{v_j \in V} P(a_1, a_2 \dots a_n | v_j) P(v_j) \end{aligned}$$

$$\begin{aligned} h_{MAP} &\equiv \operatorname{argmax}_{h \in H} P(h | D) \\ &= \operatorname{argmax}_{h \in H} \frac{P(D | h) P(h)}{P(D)} \\ &= \operatorname{argmax}_{h \in H} P(D | h) P(h) \end{aligned}$$

- Naïve Bayes Classifier: $\mathbf{v}_{NB} = \operatorname{argmax}_{v_j \in V} p(v_j) \prod_i p(a_i | v_j)$

Naïve Bayes Classifier Algorithm - Example

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Naïve Bayes Classifier Algorithm - Example

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- New instance to be classified

*< Outlook = Sunny, Temp = Cool,
Humidity = High, Wind = Strong >*

- Naïve Bayes Classifier

- $$v_{NB} = \underset{v_j \in V}{\operatorname{argmax}} p(v_j) \prod_i p(a_i | v_j)$$

$$p(\text{yes} | \text{new Instance}) = p(\text{yes}) * p(\text{sunny} | \text{yes}) * p(\text{cool} | \text{yes}) * p(\text{high} | \text{yes}) * p(\text{strong} | \text{yes})$$

$$p(\text{no} | \text{new Instance}) = p(\text{no}) * p(\text{sunny} | \text{no}) * p(\text{cool} | \text{no}) * p(\text{high} | \text{no}) * p(\text{strong} | \text{no})$$

Naïve Bayes Classifier Algorithm - Example

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- $p(\text{yes}) = \frac{9}{14}$ and $p(\text{no}) = \frac{5}{14}$

Outlook	Yes	No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rainy	3/9	3/5

Temp	Yes	No
Hot	2/9	3/5
Mild	4/9	2/5
Cool	3/9	2/5

Humidity	Yes	No
High	3/9	4/5
Normal	6/9	1/5

Wind	Yes	No
Strong	3/9	3/5
Weak	6/9	2/5

$$p(\text{yes}|\text{new Instance}) = p(\text{yes}) * p(\text{sunny}|\text{yes}) * p(\text{cool}|\text{yes}) * p(\text{high}|\text{yes}) * p(\text{strong}|\text{yes})$$

$$p(\text{no}|\text{new Instance}) = p(\text{no}) * p(\text{sunny}|\text{no}) * p(\text{cool}|\text{no}) * p(\text{high}|\text{no}) * p(\text{strong}|\text{no})$$

Naïve Bayes Classifier Algorithm - Example

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- $p(\text{yes}) = \frac{9}{14}$ and $p(\text{no}) = \frac{5}{14}$

Outlook	Yes	No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rainy	3/9	3/5

Temp	Yes	No
Hot	2/9	3/5
Mild	4/9	2/5
Cool	3/9	2/5

Humidity	Yes	No
High	3/9	4/5
Normal	6/9	1/5

Wind	Yes	No
Strong	3/9	3/5
Weak	6/9	2/5

PlayTennis = No

$$p(\text{yes}|\text{new Instance}) = p(\text{yes}) * p(\text{sunny}|\text{yes}) * p(\text{cool}|\text{yes}) * p(\text{high}|\text{yes}) * p(\text{strong}|\text{yes}) = \mathbf{0.0053}$$

$$p(\text{no}|\text{new Instance}) = p(\text{no}) * p(\text{sunny}|\text{no}) * p(\text{cool}|\text{no}) * p(\text{high}|\text{no}) * p(\text{strong}|\text{no}) = \mathbf{0.0206}$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$\langle \text{Outlook} = \text{sunny}, \text{Temperature} = \text{cool}, \text{Humidity} = \text{high}, \text{Wind} = \text{strong} \rangle$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$$

$$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$$

Outlook	Y	N		Humidity	Y	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		Strong	3/9	3/5
mild	4/9	2/5		Weak	6/9	2/5
cool	3/9	1/5				

$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$

$$v_{NB} = \operatorname{argmax}_{v_j \in \{yes, no\}} P(v_j) \prod_i P(a_i | v_j)$$

$$= \operatorname{argmax}_{v_j \in \{yes, no\}} P(v_j) \quad P(Outlook = sunny | v_j) P(Temperature = cool | v_j) \\ \cdot P(Humidity = high | v_j) P(Wind = strong | v_j)$$

$$v_{NB}(yes) = P(yes) P(sunny|yes) P(cool|yes) P(high|yes) P(strong|yes) = .0053$$

$$v_{NB}(no) = P(no) P(sunny|no) P(cool|no) P(high|no) P(strong|no) = .0206$$

$$v_{NB}(yes) = \frac{v_{NB}(yes)}{v_{NB}(yes) + v_{NB}(no)} = 0.205$$

$$v_{NB}(no) = \frac{v_{NB}(no)}{v_{NB}(yes) + v_{NB}(no)} = 0.795$$

- Estimate conditional probabilities of each attributes {color, legs, height, smelly} for the species classes: {M, H} using the data given in the table.
- Using these probabilities estimate the probability values for the new instance – (Color=Green, legs=2, Height=Tall, and Smelly=No).

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

New Instance

(Color=Green, legs=2, Height=Tall, and Smelly=No)

$$P(M) = \frac{4}{8} = 0.5 \quad P(H) = \frac{4}{8} = 0.5$$

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	4/4
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4

$$P(M) = \frac{4}{8} = 0.5 \quad P(H) = \frac{4}{8} = 0.5$$

Color	M	H
White	2/4	3/4
Green	2/4	1/4

Legs	M	H
2	1/4	4/4
3	3/4	0/4

Height	M	H
Tall	3/4	2/4
Short	1/4	2/4

Smelly	M	H
Yes	3/4	1/4
No	1/4	3/4

$$p(M|New Instance) = p(M) * p(Color = Green|M) * p(Legs = 2|M) * p(Height = tall|M) * p(Smelly = no |M)$$

$$p(M|New Instance) = 0.5 * \frac{2}{4} * \frac{1}{4} * \frac{3}{4} * \frac{1}{4} = 0.0117$$

$$p(H|New Instance) = p(H) * p(Color = Green|H) * p(Legs = 2|H) * p(Height = tall|H) * p(Smelly = no |H)$$

$$p(H|New Instance) = 0.5 * \frac{1}{4} * \frac{4}{4} * \frac{2}{4} * \frac{3}{4} = 0.047$$

$$p(H|New Instance) > p(M|New Instance)$$

Hence the new instance belongs to Speices H

Advantages

- It is easy and fast to predict the class of the test data set.
- It also performs well in multi-class prediction.
- When assumption of independence holds, a Naive Bayes classifier performs better compare to other models like logistic regression and you need less training data.
- It perform well in case of categorical input variables compared to numerical variable(s). For numerical variable, normal distribution is assumed (bell curve, which is a strong assumption).

Disadvantages

- If categorical variable has a category (in test data set), which was not observed in training data set, then model will assign a 0 (zero) probability and will be unable to make a prediction. This is often known as Zero Frequency. To solve this, we can use the smoothing technique. One of the simplest smoothing techniques is called Laplace estimation.
- On the other side naive Bayes is also known as a bad estimator, so the probability outputs are not to be taken too seriously.
- Another limitation of Naive Bayes is the assumption of independent predictors. In real life, it is almost impossible that we get a set of predictors which are completely independent.

Applications

- **Real time Prediction:** Naive Bayes is an eager learning classifier and it is sure fast. Thus, it could be used for making predictions in real time.
- **Multi class Prediction:** This algorithm is also well known for multi class prediction feature. Here we can predict the probability of multiple classes of target variable.
- **Text classification/ Spam Filtering/ Sentiment Analysis:** Naive Bayes classifiers mostly used in text classification (due to better result in multi class problems and independence rule) have higher success rate as compared to other algorithms. As a result, it is widely used in Spam filtering (identify spam e-mail) and Sentiment Analysis (in social media analysis, to identify positive and negative customer sentiments)
- **Recommendation System:** Naive Bayes Classifier and Collaborative Filtering together builds a Recommendation System that uses machine learning and data mining techniques to filter unseen information and predict whether a user would like a given resource or not.